

Subject Description Form

Subject Code	SFT405FI
Subject Title	Advanced Performance Evaluation of Functional Apparel
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: SFT202FY Introduction to Fashion Materials and SFT204FY Introduction to Apparel Techniques Exclusion: ITC4201T Advanced Performance Evaluation of Functional Apparel
Objectives	The subject provides advanced technical knowledge and skills for the performance evaluation of various functional fabrics and apparel products through applying effective forms of assessment techniques and analysis methods. It identifies the problems encountered in the performance evaluation and analysis of functional apparel under industrial practice and addresses the solutions to satisfy the customers and industrial demands.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) understand techniques and methods for analyzing functional performance of technical fabrics to meet customers' demands. (b) design and apply subjective evaluation tools and methods to assess biosensory responses towards functional fabrics and apparel. (c) analyze wearing performance of functional apparel used for various occasions or end uses by applying rational solutions and/or experimental protocols. (d) develop capabilities to estimate appropriateness of the applied evaluation methods, thus facilitating product design and decision making of functional apparel.
Subject Synopsis/ Indicative Syllabus	(I) Instrumental Evaluation of Functional Fabrics Fabric performance-centered evaluation methods and techniques in terms of physical, mechanical, and biological properties of functional fabrics (e.g., wicking, quick dry, thermal regulation, anti-static, anti-bacterial, UV protection, smart sensing, and emerging or advanced functional materials, etc); working principle and applications of related advanced instruments and evaluation system.

	(II) Subjective/Biosensory Evaluation of Functional Apparel Neurophysiological-based biosensory assessment, fitting satisfactory assessment, thermal/mechanical and comfort performance assessment by applying rational instruments/manikins, scaling tools and experimental trials. (III) Wearing Performance Evaluation of Functional Apparel Global assessment of wearing performance of functional apparel. Multiple-evaluation techniques that simulate wearing scenarios to be introduced to enhance the understanding of psycho-physiological characteristics or biomechanical performances of various functional apparel (active, protective, healthcare, or medical, etc). (IV) Application of Multidisciplinary Approaches Motion capture test, pressure evaluation, ergonomic analysis, and wear trial, etc., for assessment of functional apparel designed for specific users.																																																						
Teaching/Learning Methodology	Lectures will be structured to convey theories and concepts, with specific case studies involving emerging functional apparel products. Tutorials will be designed to deepen the knowledge and skills delivered in lectures and to provide a laboratory environment for practice and discussion. A group project, presentations and written reports will be employed for assessment of critical thinking, analysis capabilities, problem solving, teamwork and communication skills.																																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th></th><th></th></tr><tr><td>Continuous Assessment</td><td>100%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td><i>1. Lab Report</i></td><td><i>20%</i></td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td><i>2. Group Project</i></td><td><i>30%</i></td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td><i>3. Individual Report</i></td><td><i>50%</i></td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table>	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d			Continuous Assessment	100%	✓	✓	✓	✓			<i>1. Lab Report</i>	<i>20%</i>	✓	✓	✓	✓			<i>2. Group Project</i>	<i>30%</i>	✓	✓	✓	✓			<i>3. Individual Report</i>	<i>50%</i>	✓	✓	✓	✓			Total	100%						
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Assessment Methods in Alignment with Intended Learning Outcomes	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: This subject will provide students with lectures and laboratory training. Lab report will be used to evaluate students' abilities to identify and describe the performance evaluation methodologies, process and techniques. A group project and report will assess students' knowledge of the methods appropriate to the performance and properties of functional fabrics and apparel. An individual report will be used to assess students' knowledge of theory, concepts, principles and appropriate adoption of the methodologies used in functional apparel evaluation and analysis.	
Student Study Effort Expected	Class contact:	
	Lecture	21 Hrs.
	Laboratory	18 Hrs.
	Other student study effort:	
	Assignments	30 Hrs.
	Reading and research	40 Hrs.
	Total student study effort	109 Hrs.
Reading List and References	<u>Books</u> Majumdar, A., Gupta, D., Gupta, S. (2020). <i>Functional Textiles and Clothing</i>. Singapore: Springer. Zakaria, N. and Gupta, D. (2019). <i>Anthropometry, Apparel Sizing and Design</i>. San Diego: Elsevier Science & Technology. Hayes, S.G. and Venkatraman, P. (2018). <i>Materials and Technology for Sportswear and Performance Apparel</i>. Boca Raton: CRC Press. Nayak, R. and Padhye, R. (2017). <i>Manikins for Textiles Evaluation</i>. Kidlington, United Kingdom: Woodhead Publishing, an imprint of Elsevier. McLoughlin, J. and Sabir, T. (2017). <i>High-Performance Apparel: Materials, Development, and Applications</i>. Cambridge: Elsevier Science & Technology.	

	Scherer, W. (2016). <i>Advances in Women's Intimate Apparel Technology</i>. Amsterdam, Netherlands: Woodhead Publishing. Brown, P. and Janett, R. (2014). <i>Ready-to-wear Apparel Analysis</i>. Boston: Pearson. Das, A. and Alagirusamy, R. (2010). <i>Science in Clothing Comfort</i>. Cambridge: Woodhead Publishing India PVT. LTD. Fabbri, K. (2015), <i>Indoor Thermal Comfort Perception</i>. Springer. Watkins, S.M. and Lucy E.D. (2015). <i>Functional Clothing Design: from Sportswear to Spacesuits</i>. New York: Fairchild Books, an imprint of Bloomsbury Publishing Inc.
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